Joey D. and the Floating Geese

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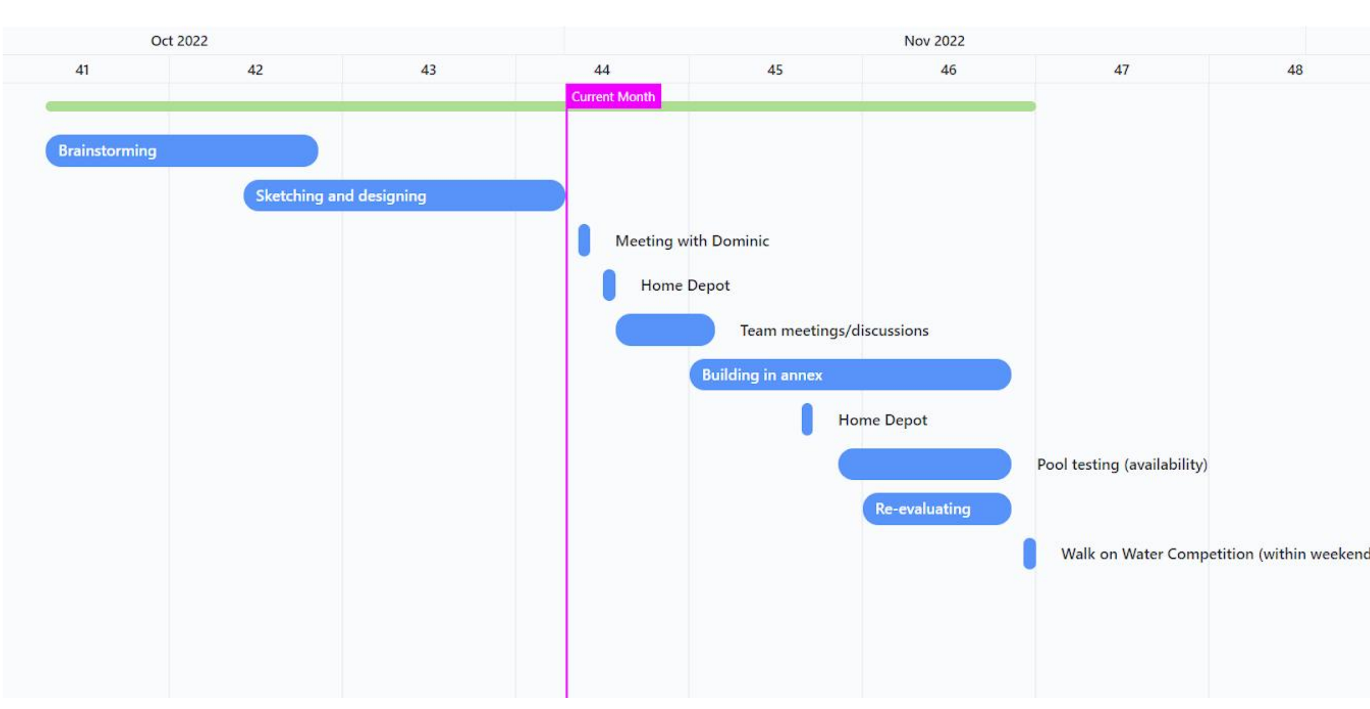
ABSTRACT

For our Walk on Water Project, our goal was to design and build a boat that can get a member of our team across the length of the pool without falling in the water. Over the past two months, our team has been meeting, designing, and creating a boat that will successfully carry one of us across the pool.

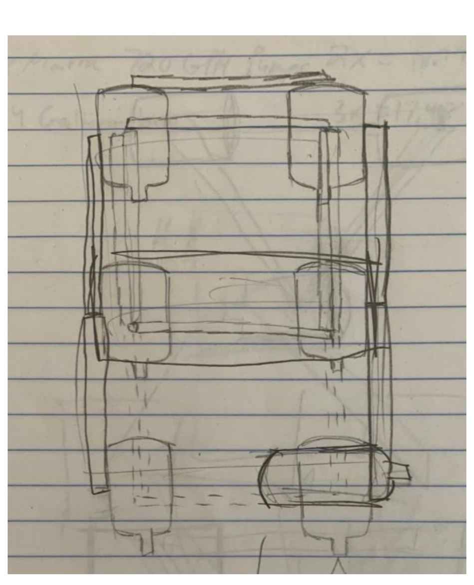
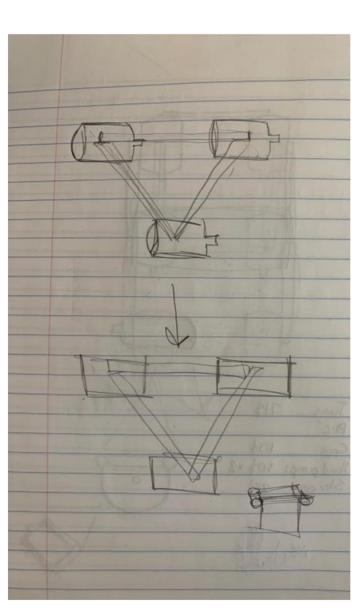
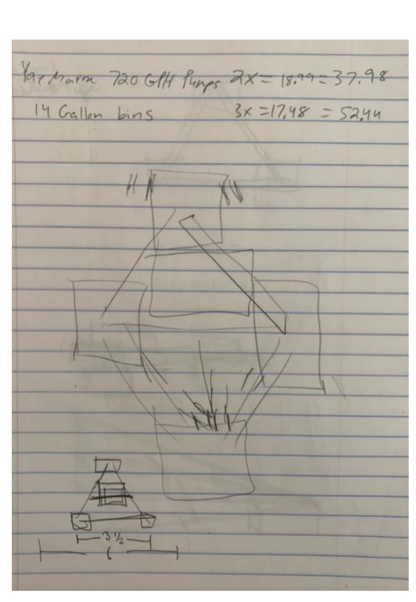
Problem Definition

Our group was asked to build a strong structure that would successfully be able to get across the pool with a team member onboard. Our team budget was $200, so we had to make sure that our materials were not too expensive while also making sure we had enough materials. In terms of our schedules, it was difficult to find times to meet because of our class times and we wanted to make sure that we all would be able to meet. This was especially difficult when we were planning to go to Home Depot to get materials. In addition, were faced with redesigning our structure to assure that it would get across the pool. Our team used a Gantt Chart to help us plan our project and stay on track.



Concept Development

Our boat went through several designs which included designs for the boat frame, ways to keep our boat afloat, and finally meeting with the head of the machine shops to finalize our design.

Detail Design

- Three airtight 12-gallon bins shaped in a triangle as our base
- Four cut 2.0” PVC pipes to hold our bins together
- Two cut 1.5” PVC pipes as a frame for our operator to hold onto
- Duct tape to tape our bin and PVC together
- Screws to keep our PVC secure

Testing and Refinement

To test our design, we went to the pool to see if we would be able to use it. The boat was able to float with a person on it, however, it tipped when we tried to propel ourselves forward. We then modified our design and shortened the handles which was successful

Analysis of Design Process

By using the design process, our team was able to schedule our time, design and test our boat, and then re-evaluate and re-design based on our observations.